

Arturo Alcorta Sánchez

JUNIOR AI / COMPUTER VISION ENGINEER

🏠 ArturoAlcorta.github.io | 📄 ArturoAlcorta | 🌐 ArturoAlcorta

Summary

Junior AI & Computer Vision Engineer combining a solid foundation in deep learning with hands-on experience in embedded systems and sensor integration. Proven track record in R&D at Técnicas Reunidas developing and deploying end-to-end vision solutions in industrial environments. Passionate about robotics and bridging the gap between raw physical hardware and AI perception stacks. Author of peer-reviewed scientific publications with the ability to take complex projects from prototype to deployment. Cambridge C2 Proficiency, eager to tackle applied challenges in autonomous systems.

Professional & Research Experience

Técnicas Reunidas

Madrid, Spain

COMPUTER VISION ENGINEER · AI R&D DEPARTMENT

September 2025 – Present

- **AI-Based Applications:** LLM-powered tools and solutions for internal process automation and acceleration.
- **Computer Vision Models:** Training and fine-tuning of vision models (**YOLO**) for object detection in construction site surveillance.
- **MCP Server:** MCP (**FastMCP**) server that exposes applications to a corporate internal LLM, enabling non-technical employees to interact with them in natural language.
- **Stack:** FastAPI, Celery, PostgreSQL, Docker, YOLO, Agentic Systems, FastMCP, HTMX.

IQM - CSIC – Prof. José Elguero

Madrid, Spain

VOLUNTARY COLLABORATION IN AI APPLIED TO SCIENTIFIC RESEARCH

January 2025 – Present

- **AI Researcher:** Co-authored 2 scientific publications applying ML models for data analysis and workflow automation in scientific research.
- **Stack:** XGBoost, Neural Networks, Bayesian Optimization, Optuna

Personal Projects

VBT tracker

MONOCULAR-VISION VELOCITY ENCODER (VBT) — SOFTWARE ALTERNATIVE TO COMMERCIAL DEVICES

ArturoAlcorta/VBT-tracker

- **Metric velocity from a single camera:** Real bar velocity and displacement using the standard plate diameter as scale, with no dedicated hardware.
- **Validation:** Velocity error of 0.05 m/s against approximate measurements; sticking-point detection and RIR estimation from intra-set velocity loss (grounded in strength-training literature), qualitatively validated with 6 users.
- **Stack:** Ultralytics/YOLO (mAP50 0.995, single class; dataset annotated in CVAT with SAM assistance), SAM, CVAT, NumPy.

Gym technique analyzer

MONOCULAR-VISION LIFTING TECHNIQUE ANALYSIS — PER-REPETITION FEEDBACK

ArturoAlcorta/Gym-technique-analysis

- **Per-repetition analysis with no hardware:** 2D pose with SynthPose (ViTPose-H, 52 keypoints) on top of YOLO11m detection; 3D lifting was discarded as it deformed the limbs by up to 70% within a single repetition; rep counting via a 4-state hysteresis and per-rep biomechanical metrics (valgus, thoracolumbar flexion, hip-knee phase).
- **Scoring:** Multivariate DTW against reference repetitions and relational metrics invariant to body proportions; also operates without a comparison. Metrics defined together with a professional trainer.
- **Stack:** PyTorch, Transformers, Ultralytics, DTW.

Education

MSc in Deep Learning for Audio and Video Signal Processing (MUDL4AVSP)

Madrid, Spain

UNIVERSIDAD AUTÓNOMA DE MADRID (UAM)

September 2024 – July 2025

- Advanced training in deep neural networks applied to audiovisual signal processing.
- **Master's Thesis — Exploring Low-Rank Adaptation for Generating Synthetic Images in Autonomous Driving Datasets:** Implementation and fine-tuning of LoRA on diffusion models to generate synthetic images of complex urban environments; 70% reduction in training time and 99.9% in trainable parameters, while improving image quality.
- **Stack:** Python, PyTorch, TensorFlow, Hugging Face Diffusers, LoRA, Stable Diffusion, Transformers, LLMs, MLflow.

BSc in Telecommunication Technologies and Services Engineering

Madrid, Spain

UNIVERSIDAD AUTÓNOMA DE MADRID (UAM)

September 2019 – July 2024

- **Bachelor's Thesis:** Portable air quality monitoring system integrating low-cost environmental sensors with a Raspberry Pi Zero.
- Development and deployment of machine learning models to calibrate the integrated sensors, ensuring accurate data acquisition under real-world conditions.
- **Stack:** MATLAB, C, C++, Python, Embedded Systems, Signal Processing, Machine Learning, Docker, FPGAs.